

SPEC FLOW AI

Revised App Flow Document

v1 Notion Product Context Auditor + Reviewable Task Generator

CEO + Senior PM + Senior SWE Version · May 2026 · Confidential

North Star

SpecFlow v1 must be boringly reliable before it becomes impressively agentic. The product wins only if users trust that it read the right sources, cited claims correctly, preserved their edits, and created Notion tasks without duplicating or polluting the workspace.

0. Executive Summary

One-line v1

Selected Notion context -> cited product audit -> persistent task review -> approved Notion dispatch.

- The previous App Flow had a solid structure, but made unsafe v1 tradeoffs: local-only task review state, 500-page source limits, immediate write permissions, weak deduplication, simplistic retrieval scoring, and missing idempotency.
- This revised flow narrows the product around trust: read-only audit first, explicit source selection, cited claims, persisted review state, and idempotent Notion dispatch.
- The v1 product is not an autonomous PM. It is a trusted product context auditor that helps founders and PMs find decisions, open questions, contradictions, and next tasks from messy Notion workspaces.

1. Product Principles

Principle	Meaning in the Product	Engineering Implication
Trust before automation	User must see sources before accepting output.	Every claim and task needs citations. No uncited claims.
Review before write	Nothing touches Notion until user approves.	Dispatch endpoint only accepts approved server-side task IDs.
Read-only first	Do not ask for write permissions until dispatch.	Progressive Notion permission validation.
Small source set first	Quality beats workspace-wide ingestion.	100-page hard cap, 30-page warning, source suggestions.
Idempotent writes	Retries must not duplicate Notion tasks.	Store Notion page IDs and use idempotency keys.
Observable AI	AI output must be debuggable.	Persist model, prompt version, retrieval version, chunk IDs, validation failures.

2. Revised Screen Inventory

ID	Screen	Route	Core Job
S-01	Landing / Connect	/ or /connect	Explain value and connect Notion read-only.
S-02	OAuth Callback	/auth/notion/callback	Handle Notion OAuth result. No UI except fallback.
S-03	Source Picker	/setup/sources	Search, suggested sources, exclusions, source-of-truth markers.
S-04	Target DB Selector	/setup/target-db	Optional until dispatch; choose task DB or skip.
S-05	Ingestion Status	/ingestion/[runId]	Live ingestion feed, skipped pages, source health.
S-06	Dashboard	/dashboard	Trust health, last audit, stale docs, next action.
S-07	Analysis Loading	/analysis/[analysisId]/loading	Streaming progress.
S-08	Audit Report	/analysis/[analysisId]/report	Critical contradictions, open questions, decisions, suggested tasks.
S-09	Contradiction Detail	/analysis/[analysisId]/contradiction/[id]	Side-by-side evidence and proposed resolution task.

S-10	Task Review	/analysis/[analysisId]/tasks	Persistent approve/edit/reject workflow.
S-11	Permission Upgrade	/analysis/[analysisId]/permissions	Request Notion write permission only if needed.
S-12	Dispatch Confirmation	/analysis/[analysisId]/dispatch	Show created tasks and Notion links.
S-13	Run History	/history	Past audits, feedback, accepted/rejected task metrics.
S-14	Settings	/settings	Sources, exclusions, target DB, connection health.
S-15	Error	/error	Safe generic fallback with recovery path.
S-16	Internal Run Debug	/internal/runs/[analysisId]	Admin-only AI/debug observability.

3. Flow A - Onboarding and Notion Connection

Change from original

Ask for read access first. Do not request write permissions until the user chooses to dispatch approved tasks.

Step	Type	User/System Action	PM Requirement	Dev Requirement
A1	SCREEN	S-01 Landing: Connect Notion	Single CTA. Trust copy: "We only read pages you select."	Check Clerk session. If configured, redirect to dashboard.
A2	ACTION	User clicks Connect Notion	No forms. No feature wall.	Start Notion OAuth with read-only access where possible. Store PKCE verifier.
A3	API	Notion OAuth callback	If denied, give exact reason and retry.	Exchange code, encrypt token, store connection row. Never log token.
A4	DECISION	Connection success?	Success -> source picker. Failure -> retry with clear message.	Set notionConnected=true. Log oauth_error if failed.
A5	SCREEN	S-03 Source Picker	Search-first UX with suggested product sources and exclusions.	Fetch Notion pages/databases. Cache results. Apply 100-page hard cap.
A6	ACTION	User selects sources	Show selected count, estimated time, warnings above 30 pages.	Save selected page/database IDs. Mark source-of-truth and excluded IDs.
A7	DECISION	Zero sources selected?	Disable Continue. Inline message: "Select at least one source."	Client + server validation.
A8	SCREEN	Optional target DB selector	Explain: "You can pick this now or at dispatch."	Fetch writable databases if write token exists; otherwise defer.
A9	ACTION	Start ingestion	Confirmation summary: selected sources, excluded areas, estimated duration.	POST /api/ingestion/start -> enqueue Inngest job -> return runId.

4. Source Picker Requirements

Area	Required Behavior
Suggested product sources	Auto-detect likely PRDs, roadmaps, decision logs, interview databases, feedback docs, and meeting notes. Show reason for suggestion.
Search	Search by page title, database title, detected type, and recently edited date.
Exclusions	User can exclude HR, legal, finance, personal, archived, and irrelevant spaces.
Source-of-truth marker	User can mark key pages/databases as authoritative. Retrieval scoring must boost these.
Limits	Soft warning above 30 sources. Hard cap at 100 pages/databases for v1.
Preview	Show top-level titles and detected type before ingestion. No full workspace blind crawl.

```
type SuggestedSource = {
  pageId: string;
  title: string;
```

```

detectedType: "prd" | "roadmap" | "decision_log" | "feedback" | "meeting_note" | "unknown";
confidence: "high" | "medium" | "low";
reason: string;
lastEditedAt: string;
userMarkedSourceOfTruth?: boolean;
};

```

5. Flow B - Ingestion Pipeline

Step	Type	System Behavior	PM Requirement	Dev Requirement
B1	SCREEN	S-05 live ingestion feed	Show queued, processed, skipped, failed, duplicates, stale docs.	SSE preferred. Polling fallback if SSE disconnects repeatedly.
B2	SYSTEM	Fetch selected sources only	User should see no surprise data access.	Respect selected source IDs and exclusions. Rate-limit Notion calls.
B3	SYSTEM	Parse Notion block tree	Extract readable content and source path.	Handle paragraphs, headings, lists, toggles, callouts, code, tables, quotes. Unknown -> text fallback.
B4	SYSTEM	Classify chunks	Show detected type badges in summary.	Assign sourceType and authorityRank.
B5	SYSTEM	Deduplicate and version detect	Tell user when duplicates/stale docs are skipped or downgraded.	Exact hash, near-duplicate similarity, version-title detection.
B6	SYSTEM	Embed and store	Show "Indexing context..." state.	Batch embeddings. Store vectors in pgvector.
B7	SYSTEM	Finalize source health	Summary: active, stale, duplicate, failed, permission-lost.	Upsert ingestion_runs and context_chunks.
B8	DECISION	Failure threshold	If >=10% failed, allow proceed with prominent warning.	Set PARTIAL_FAILURE and list failed pages.
B9	ERROR	Total ingestion failure	Message: "Your Notion pages were not modified."	Clear partial chunks for runId or mark them failed.

6. Context Chunk Lifecycle

Required correction

The system must never analyze deleted, stale, permission-lost, duplicate, or excluded chunks as if they were active context.

```

type ContextChunkStatus =
  | "active"
  | "stale"
  | "deleted"
  | "permission_lost"
  | "duplicate"
  | "failed";

type ContextChunk = {
  id: string;
  workspaceId: string;
  source: "notion";
  pageId: string;
  blockId?: string;
  title: string;

```

```
content: string;
contentHash: string;
sourceType: "prd" | "roadmap" | "decision_log" | "feedback" | "meeting_note" | "unknown";
authorityRank: number;
status: ContextChunkStatus;
userMarkedSourceOfTruth: boolean;
lastSeenAt: string;
sourceLastEditedAt: string;
pageUrl: string;
blockUrl?: string;
embedding?: number[];
};
```

7. Flow C - Run Analysis and Audit Report

Step	Type	Action	PM Requirement	Dev Requirement
C1	ACTION	User clicks Run Audit	Button disabled unless ingestion is complete or partial with warning accepted.	POST /api/analysis/start. Validate workspace ownership and active chunks.
C2	SCREEN	S-07 analysis loading	Show stages: retrieving context, checking contradictions, validating claims, preparing report.	SSE events: RETRIEVAL_START, SYNTHESIS_START, VALIDATION_DONE, COMPLETE.
C3	SYSTEM	Ranked retrieval	Use source authority, recency, source-of-truth, and semantic similarity.	Filter out stale/deleted/permission-lost/duplicate chunks.
C4	SYSTEM	Claude synthesis	Evidence-first prompt. No claim without sources.	Persist model, promptVersion, retrievalVersion, token usage.
C5	SYSTEM	Claim validation	Remove unsupported claims and show warning if many removed.	Validate citations, quote existence, freshness, contradiction source count.
C6	SCREEN	S-08 audit report	Prioritize critical contradictions, open decisions, decisions made, suggested tasks.	Store claims/tasks server-side. Render citations inline.
C7	ACTION	User opens citation	Trust moment. Exact excerpt visible even if Notion link fails.	Resolve block URL with page fallback. Track citation_clicked.
C8	ACTION	User opens contradiction	Side-by-side sources, why it matters, suggested resolution.	No new model call. Use stored contradiction object.
C9	ACTION	User reviews tasks	Tasks are already persisted and editable.	Navigate to S-10. Fetch task_suggestions by analysisId.

8. Retrieval Scoring

```
retrieval_score =
  0.45 * semantic_similarity
+ 0.20 * source_authority
+ 0.15 * recency_score
+ 0.10 * user_selected_priority
+ 0.05 * citation_density
+ 0.05 * past_acceptance_signal
```

Source Type	Authority Weight
PRD	1.00
Roadmap	0.90
Decision log	0.90

User interview	0.75
Customer feedback	0.70
Meeting note	0.60
Random / unknown note	0.30

9. Audit Output Schema

```

type AuditClaim = {
  id: string;
  analysisId: string;
  claimType: "decision" | "open_question" | "contradiction" | "risk";
  statement: string;
  confidence: "high" | "medium" | "low";
  severity?: "blocking" | "important" | "minor";
  sourceFreshness: "fresh" | "possibly_stale" | "stale";
  evidence: Evidence[];
  validationStatus: "passed" | "needs_review" | "rejected";
};

```

```

type Evidence = {
  pageId: string;
  blockId?: string;
  pageTitle: string;
  quote: string;
  pageUrl: string;
  blockUrl?: string;
  fallbackUrl: string;
};

```

Validation rule

No evidence = no claim. Contradictions require at least two cited sources. A claim cannot be HIGH confidence if all evidence is stale or possibly stale.

10. Flow D - Persistent Task Review

Critical correction

Task review state must be server-persisted in v1. Local-only approval/edit/reject state is not acceptable because this is the highest-value workflow.

Step	Type	Action	PM Requirement	Dev Requirement
D1	SCREEN	S-10 task review	5-10 task cards. Sort: blocking/high confidence first.	GET /api/tasks?analysisId. Render persisted task_suggestions.
D2	ACTION	User expands task evidence	Show source excerpts and links.	Evidence JSON stored on task row. No model call.
D3	ACTION	User edits task	Inline edit title and description. Preserve original and diff.	PATCH /api/tasks/[taskId]. Optimistic UI with rollback.
D4	ACTION	User approves task	Card marked approved. Count updates.	PATCH status=approved. Persist immediately.
D5	ACTION	User rejects task	Card collapses but restore available.	PATCH status=rejected. Persist for learning.

D6	ACTION	User exports draft	Copy markdown or download markdown without Notion write.	GET /api/analysis/[id]/export.md.
D7	ACTION	User dispatches approved tasks	Confirmation modal. "Create X tasks in Notion."	POST /api/dispatch/tasks with approved server-side task IDs and idempotencyKey.

type TaskSuggestionStatus =

| "pending"
 | "approved"
 | "edited"
 | "rejected"
 | "dispatching"
 | "dispatched"
 | "dispatch_failed";

type TaskSuggestion = {

id: string;
 analysisId: string;
 workspaceId: string;
 userId: string;
 originalTitle: string;
 originalDescription: string;
 editedTitle?: string;
 editedDescription?: string;
 priority: "high" | "medium" | "low";
 confidence: "high" | "medium" | "low";
 status: TaskSuggestionStatus;
 evidence: Evidence[];
 notionPageId?: string;
 notionPageUrl?: string;
 };

11. Flow E - Permission Upgrade and Idempotent Dispatch

Step	Type	Action	PM Requirement	Dev Requirement
E1	DECISION	Does user have Notion write access?	If not, explain why write is needed only now.	Validate token scopes before dispatch.
E2	SCREEN	S-11 permission upgrade	Copy: "SpecFlow needs write access to create approved tasks only."	Restart OAuth with write permission if needed. Preserve approved task state.
E3	ACTION	User confirms dispatch	Clear statement: "This creates tasks in [DB]."	Client sends approved task IDs and idempotencyKey only.
E4	SYSTEM	Server validates dispatch	Avoid unauthorized or tampered task content.	Verify user owns workspace, task IDs belong to analysis, statuses approved/edited.
E5	SYSTEM	Create Notion tasks idempotently	Retries must not create duplicates.	If task has notionPageId, return existing URL. Else create and store ID.
E6	ERROR	Partial dispatch failure	Show succeeded and failed tasks. Per-task retry.	Failed tasks remain approved/dispatch_failed. Retry uses same task IDs.
E7	SCREEN	S-12 confirmation	Show created tasks with Notion links and "Was this	Persist dispatch_logs and update task_suggestions.

			useful?" feedback.	
--	--	--	--------------------	--

POST /api/dispatch/tasks

```
{
  analysisId: string;
  approvedTaskIds: string[];
  idempotencyKey: string;
}
```

Server rule:

- If task_suggestion.notionPageId exists, return existing URL.
- Else create Notion page, store notionPageId, notionPageUrl, and dispatch log.
- Unique index on dispatched_tasks(taskSuggestionId).

12. Flow F - Returning User and Dashboard

Dashboard Card	What It Shows	Why It Matters
Connection health	Healthy / expired / missing write permission	Prevents surprise failures before analysis or dispatch.
Active sources	Number of active selected pages/databases	Makes scope visible.
Last ingestion	Timestamp and stale warning	Prevents stale audit confusion.
Stale docs	Count and list of stale/permission-lost/deleted docs	Shows context quality.
Last audit summary	Contradictions, open questions, tasks approved	Shows value quickly.
Task acceptance rate	Approved / generated over recent runs	Quality signal for product and user.
Primary CTA	Run New Audit or Re-ingest Context	Guides user to correct next action.

13. Minimum Data Model

Table	Purpose
users	Application user profile, linked to Clerk user ID.
workspaces	SpecFlow workspace abstraction, even if v1 is single-user.
workspace_members	Future-proof team support. v1 can have one owner row.
notion_connections	Encrypted Notion token, scope state, connection health.
workspace_sources	Selected sources, excluded sources, source-of-truth markers.
ingestion_runs	Ingestion status, counts, timing, failure reasons.
context_chunks	Normalized Notion chunks, lifecycle status, embeddings, metadata.
analysis_runs	Model/prompt/retrieval versions, audit snapshots, quality metrics.
audit_claims	Validated decisions, questions, contradictions, risks.
task_suggestions	Persisted generated tasks and review state.
dispatch_logs	Notion write attempts, idempotency, successes/failures.
feedback_events	User ratings and product quality signals.

14. Revised API Endpoint Reference

Method	Endpoint	Body / Params	Response / Notes
POST	/api/auth/notion/callback	code, state	Stores encrypted token. No token logs.
GET	/api/notion/sources	workspaceId	SuggestedSource[]; search/filter supported.
POST	/api/workspace/sources	selectedIds[], excludedIds[], sourceOfTruthIds[]	Saves source set. Enforces ownership.
POST	/api/ingestion/start	workspaceId	{runId}; uses saved source set.
GET	/api/ingestion/{runId}/stream	-	SSE status events. Polling fallback endpoint allowed.
POST	/api/analysis/start	workspaceId, runId?	{analysisId}; validates active chunks.
GET	/api/analysis/{analysisId}/stream	-	SSE analysis events.
GET	/api/analysis/{analysisId}/report	-	Audit claims, contradictions, task summary.
GET	/api/tasks	analysisId	Persisted TaskSuggestion[].
PATCH	/api/tasks/{taskId}	status, editedTitle, editedDescription	Persists review state.
GET	/api/analysis/{id}/export.md	-	Markdown export fallback.

POST	/api/dispatch/tasks	analysisId, approvedTaskIds[], idempotencyKey	{succeeded[], failed[]}
GET	/api/history	workspaceId, page	Paginated AnalysisRun[]
GET	/api/dashboard	workspaceId	Trust health + summary metrics.
GET	/api/notion/block/[blockId]/deeplink	-	Block URL or page fallback.
GET	/internal/runs/[analysisId]	admin only	Prompt/retrieval/model/debug details.

15. Security and Ownership Rules

- Never trust client-provided workspaceId without checking it belongs to the authenticated user.
- Every analysis, task, source, and dispatch row must be scoped by userId and workspaceId.
- Client sends task IDs for dispatch, not task content. Server fetches canonical persisted task content.
- Notion tokens must be encrypted at rest and never logged.
- Block deeplink endpoints must verify the requested block belongs to a source in the user workspace.
- Internal debug routes must be admin-only and should redact secrets and PII by default.

```

async function requireWorkspaceAccess(userId: string, workspaceId: string) {
  const membership = await db.workspaceMembers.findFirst({
    where: { userId, workspaceId }
  });
  if (!membership) throw new AppError("WORKSPACE_ACCESS_DENIED");
  return membership;
}

```

16. Error and Empty State Reference

Code	Severity	User Message	Recovery
OAUTH_DENIED	blocking	Connection cancelled. SpecFlow needs Notion access to read selected sources.	Try again.
NO_SOURCES_SELECTED	info	Select at least one source to analyze.	Pick suggested sources or search Notion.
SOURCE_LIMIT_EXCEEDED	blocking	v1 supports up to 100 sources. Narrow your selection.	Return to source picker.
NO_PRODUCT_DOCS_DETECTED	warning	SpecFlow did not detect strong product sources in this selection.	Add PRDs, roadmaps, decision logs, or feedback docs.
INGESTION_PARTIAL_FAILURE	warning	Some pages could not be read. The audit may be incomplete.	Proceed with warning or retry.
INGESTION_FAILED	blocking	Ingestion failed. Your Notion pages were not modified.	Retry ingestion.
ANALYSIS_LOW_CONFIDENCE	warning	Some claims were removed because there was not enough evidence.	Show partial report.
NO_RELIABLE_CLAIMS	blocking	SpecFlow could not generate a reliable audit from these sources.	Re-ingest or select better sources.
NO_CONTRADICTIONS	info	No contradictions were detected in this run.	Continue to open questions and tasks.
NO_TASKS_GENERATED	info	Not enough unresolved gaps were found to create reliable tasks.	Review audit or select more sources.
WRITE_PERMISSION_REQUIRED	blocking	SpecFlow needs Notion write access to create approved tasks.	Upgrade permission.
DISPATCH_PARTIAL_FAILURE	warning	Some tasks were created. Some failed.	Retry failed tasks.
DISPATCH_DUPLICATE_PREVENTED	info	This task was already created in Notion.	Show existing Notion link.

17. Product Analytics and Evaluation

Event	Why It Matters
notion_connected	Measures onboarding start.
sources_selected	Measures setup completion and source quality.
ingestion_completed	Reliability of ingestion.
analysis_completed	Reliability and time-to-value.
citation_clicked	Trust behavior.
contradiction_opened	Core aha moment.
task_edited	Measures output quality gap.
task_approved	Measures task usefulness.
task_rejected	Negative quality signal.

dispatch_completed	Activation moment.
run_rated	Direct user feedback.
repeat_audit_run	Retention signal.

Metric	Target for v1 Beta
Connect -> first audit conversion	>= 60%
Ingestion completion rate	>= 90%
Analysis completion rate	>= 90%
Citation accuracy	>= 90%
Hallucinated claims	< 5%
Task approval rate	>= 50%
Repeat run within 7 days	>= 25%
Median time to useful report	< 2 minutes after ingestion

18. Navigation Map

From	To	Trigger	Back Behavior
S-01 Landing	S-02 OAuth	Connect Notion	N/A
S-02 OAuth	S-03 Source Picker	OAuth success	No
S-03 Source Picker	S-05 Ingestion	Sources selected + Start	Yes
S-05 Ingestion	S-07 Analysis Loading	Run Audit	No while running
S-07 Loading	S-08 Audit Report	Analysis complete	No
S-08 Audit Report	S-09 Contradiction Detail	Open contradiction	Yes
S-08 Audit Report	S-10 Task Review	Review tasks	Yes
S-10 Task Review	S-11 Permission Upgrade	Dispatch without write permission	Yes, task state preserved
S-10 Task Review	S-12 Dispatch Confirmation	Dispatch success	No for created tasks
S-12 Confirmation	S-06 Dashboard	Done	No
S-06 Dashboard	S-13 History	View history	Yes
Any authenticated	S-14 Settings	Nav/settings	Yes

19. Build Priority

Priority	Build Item	Reason
P0	Read-only Notion OAuth + source picker	Trust-first onboarding.
P0	Ingestion with chunk lifecycle states	Prevents stale/private/deleted context use.
P0	Cited audit report with validation	Core product value.
P0	Persistent task review state	Prevents loss of user edits and approvals.
P0	Idempotent dispatch	Prevents duplicate Notion tasks.
P0	Workspace ownership checks	Security baseline.
P1	Dashboard trust health	Makes returning user flow clear.
P1	Markdown export fallback	Value without write permission.
P1	Analytics and feedback events	Improves output quality.
P1	Internal debug view	Makes AI failures diagnosable.
P2	Team collaboration	Post-v1 expansion.
P2	Figma/Jira/Linear/Slack	Post-validation integrations.

20. Claude Implementation Handoff Prompt

Use this with Claude Code

Paste the prompt below with this document attached. It forces implementation from this revised flow rather than guessing from the previous build plan.

You are a senior full-stack engineer implementing SpecFlow AI v1.

Use the attached Revised App Flow Document as the source of truth.

Do not implement autonomous task dispatch. Do not use local-only task review state.

Build v1 around: read-only Notion source selection, ingestion, cited audit report, persistent approve/edit/reject task review, progressive write permission, and idempotent Notion dispatch.

Start by auditing the current repository against the revised flow. Produce:

1. Missing routes and screens.
2. Missing database tables/migrations.
3. API endpoints to add or modify.
4. Security checks needed for workspace ownership.
5. Implementation plan in small PR-sized steps.

Then implement in this order:

- P0. Workspace/source data model and ownership middleware.
- P0. Source picker with 100-source hard cap and source-of-truth markers.
- P0. Ingestion chunk lifecycle states.
- P0. Cited audit result schema and claim validation.
- P0. Persistent task_suggestions table and PATCH /api/tasks/[taskId].
- P0. Idempotent dispatch with notionPageId storage and idempotencyKey.
- P1. Dashboard trust health and product analytics events.

For every change, include validation commands, tests, and exact files changed.